

Lee Davis

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My research interests are mostly in compression schemes, especially dynamic tokenization, with the goal of finding architectures that work across modalities to align latent representations for higher quality multimodal models.

EDUCATION

BS Computer Science & Engineering

The Ohio State University

2024-2028

Columbus, OH

WORK EXPERIENCE

Research Assistant to Prof. Sachin Kumar

The Ohio State University

Feb 2025 – Present

Columbus, OH

- Lead researcher on dynamic compression for SOTA ASR models, targeting robust recognition in noisy settings.
- Iterated a dynamic tokenization module that achieved ~5% WER with 5x sequence compression on Whisper, preserving near-state-of-the-art accuracy under tight compute constraints.
- Independently diagnosed and fixed a non-convergent training regime by re-architecting the tokenization to occur pre-encoder, and proved via ablations that positional encodings were destroying performance.
- Built and maintained the entire research codebase (data pipelines, training, evaluation, ablations) while coordinating a 4-person collaboration spanning DeepMind, NVIDIA, and CMU researchers.

AI Engineer Intern

[MarginEdge](#)

May 2025 – July 2025

Ballston, VA

- Identified a knowledge bottleneck where senior staff were repeatedly answering the same questions in Slack.
- Designed and implemented a RAG system by ingesting 1,000+ public help articles into an AWS knowledge base.
- Led end-to-end experimentation (evaluation set design, prompt engineering, and model iteration) to align the assistant's behavior with users' needs, achieving a 0% error rate and sub 5% refusal rate during evaluation.

Full Stack Developer

[Andy's Pizza](#)

July 2025 – Aug 2025

Foggy Bottom, DC

- Conducted on-site observations and data collection to trace complaints about "cold pizza" to inventory age.
- Designed and built a custom RFID-based monitoring system (tagged trays + shelf readers) to log real-time pizza age and generate actionable operational data.
- Deployed across two locations and monitored results, reducing peak pizza age from ~4 hours to ~2 hours through data-driven interventions via a web-based admin dashboard with full reporting tools.

PROJECTS

[WikiChat](#) | AWS Bedrock, Firebase Auth, UX Design, Stripe Payments

Sep 2025 – Present

- Built a production web app that lets players query a complex game wiki conversationally.
- Ingested the full *Don't Starve Together* wiki into a knowledge base for accurate, fast answers.
- Integrated Firebase authentication with per-user usage tracking and rate limits to manage free and paid tiers.
- Implemented a custom Stripe subscription system to unlock higher query limits and support sustainable usage.

TECHNICAL SKILLS

Languages | Python, JavaScript, Java, C#

Systems | Firebase, AWS, Stripe, Slurm, Nvidia GPUs, TPUs

Developer Tools | Git, Visual Studio Code, Claude Code, Gemini CLI, Codex, Github Copilot

Libraries | PyTorch, Huggingface Transformers/Datasets, Nvidia NeMo, BeautifulSoup